INT396:UNSUPERVISED LEARNING

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: explain fundamental concepts of unsupervised learning and real-world problem formulations.

CO2 :: differentiate various unsupervised learning paradigms including hard and soft clustering, partition-based, hierarchical, and density-based approaches.

CO3 :: apply standard unsupervised learning algorithms such as k-Means, Hierarchical Clustering, DBSCAN, GMM, PCA, t-SNE, UMAP, and association rule mining techniques on real-world datasets using Python frameworks.

CO4 :: evaluate clustering, dimensionality reduction, and anomaly detection results using appropriate validation metrics and visualization techniques.

CO5 :: integrate multiple unsupervised learning techniques to construct visual analytics dashboards and exploratory data analysis pipelines.

CO6 :: design a comprehensive unsupervised learning project involving problem identification, data preprocessing, algorithm selection, model evaluation, ethical considerations, and effective presentation of insights.

Unit I

Foundations of Unsupervised Learning : Supervised vs. Unsupervised vs. Semi-Supervised Learning, Problem formulation for unsupervised learning, Real-life use cases: Market segmentation, Customer behavior analysis, Anomaly & fraud detection, Pattern discovery in biological and social data, Distance & similarity metrics:Euclidean, Manhattan, Cosine similarity, when and why distance choice matters

Unit II

Partition-Based Clustering : Clustering fundamentals and assumptions, Hard vs. Soft clustering, k-Means algorithm: Objective function,Initialization strategies (Random, k-Means++), Convergence and limitations, k-Medoids (PAM) vs. k-Means, Data standardization and scaling impact, MiniBatch k-Means for large-scale datasets, Cluster Validation: Inertia (WCSS), Silhouette Coefficient, Davies–Bouldin Index, Elbow method pitfalls

Unit III

Hierarchical & Density-Based Clustering : Hierarchical clustering: Agglomerative vs. Divisive, Linkage methods (single, complete, average, Ward), Dendrogram interpretation, Density-based clustering: DBSCAN fundamentals, e-neighborhood, MinPts, Noise and border points, Comparison: k-Means vs. Hierarchical vs. DBSCAN

Unit IV

Dimensionality Reduction and Representation Learning : Need for Dimensionality Reduction, Linear Dimensionality Reduction Techniques (PCA – Geometric Intuition), Manifold Learning Overview, Non-Linear Dimensionality Reduction Concepts -t-SNE, UMAP – Conceptual, Autoencoder Intuition

Unit V

Association Rule Mining & Anomaly Detection: Association Rule Mining : Support, Confidence, Lift, Conviction, Apriori Algorithm, FP-Growth algorithm, Market Basket Analysis Anomaly Detection: Anomaly vs. novelty detection, Isolation Forest, Local Outlier Factor (LOF), Applications in Cybersecurity and Fraud Detection

Unit VI

Evaluation and Applications of Unsupervised Learning : Internal and External Clustering Validation Metrics, Silhouette Score and Cohesion–Separation Intuition, Stability-Based Evaluation, Interpretability Challenges in Unsupervised Learning, Real-World Case Studies, Ethical Considerations in Pattern Discovery, Case Study : Visualizing handwritten digits (MNIST) or customer segmentation data

Project Component : Create a visualization dashboard comparing PCA vs. t-SNE results

List of Practicals / Experiments:

List of Practicals

- •Implementation and comparison of Euclidean, Manhattan, and Cosine distance metrics on a real-world dataset.

- •Comparative analysis of k-Means and k-Medoids (PAM) clustering on noisy and outlier-prone datasets.
- •Implementation of k-Means clustering from scratch and using scikit-learn.
- •Study of the impact of feature scaling and standardization on k-Means clustering results.
- •Determination of the optimal number of clusters using Elbow Method, Silhouette Score, and Davies–Bouldin Index.
- •Implementation of Hierarchical Clustering using different linkage methods and interpretation of dendrograms.
- •Implementation of DBSCAN and analysis of e (epsilon) and MinPts parameters.
- •Comparative study of k-Means, Hierarchical Clustering, and DBSCAN on the same dataset.
- •Implementation of Principal Component Analysis (PCA) and analysis of variance retention.
- •Visualization of high-dimensional data using PCA and interpretation of principal components.
- •Visualization and comparison of dimensionality reduction techniques using PCA, t-SNE, and UMAP.
- •Implementation of Apriori and FP-Growth algorithms for Market Basket Analysis and interpretation of association rules.
- •Implementation and comparison of Isolation Forest and Local Outlier Factor (LOF) for anomaly detection.
- •End-to-end Unsupervised Learning mini-project including clustering, dimensionality reduction, evaluation, and visualization dashboard.

Text Books:

1. HANDS-ON UNSUPERVISED LEARNING USING PYTHON: HOW TO BUILD APPLIED MACHINE LEARNING SOLUTIONS FROM UNLABELED DATA by ANKUR A. PATEL, O'REILLY

References:

1. KICKSTART UNSUPERVISED MACHINE LEARNING: MASTER UNSUPERVISED MACHINE LEARNING THROUGH PATTERN DISCOVERY, CLUSTERING, AND DIMENSIONALITY REDUCTION TO BUILD INTELLIGENT, REAL-WORLD APPLICATIONS by DR. NIMRITA KOUL, AVA PUBLISHING
2. MODELS AND ALGORITHMS FOR UNSUPERVISED LEARNING: PRACTICAL UNSUPERVISED MACHINE LEARNING by VAIBHAV VERDHAN, Manning Publication

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Course Outcomes: Through this course students should be able to

CO1 ::